JHU Engineering for Professionals
Applied and Computational Mathematics
Data Mining: 625.740 Fall '26

Midterm Exam

Please type up your answers. If you are not able to type your answers, please write very neatly. Sloppy, small, or otherwise difficult-to-read writing will not be accepted. Please include in your submission all code in electronic form.

You may use the textbook Pattern Classification, by Duda, et al. and other textbooks listed in the course syllabus. You may use the material posted to Canvas and Piazza: notes, videos, textbook chapters. Please cite *all* sources. No other sources may be consulted during this exam.

Posting questions to or viewing questions at "homework and exam" solving websites and the like is not allowed. Searching the internet, or obtaining notes from other classes is not allowed on this exam. During the time of the exam, you may discuss these problems only with the course instructors and nobody else.

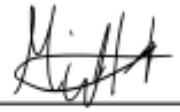
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Name: Michael Huang

Time and date of exam start: 10/19/2026 11:15 PM EDT

Time and date of exam finish: 10/22/2026 11:15 PM EDT

This exam represents my own work. I did not consult with anybody or access any prohibited sources during the time of the exam.

Signature:  **Date:** 10/22/2026

Michael Huang
EN.625.740.81
Midterm

1.

Recall that if $f(\mathbf{x})$ and $g(\mathbf{x})$ are probability distributions, then the Kullback-Leibler divergence of g from f is

$$D(f,g) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(\mathbf{x}) \log[\frac{f(\mathbf{x})}{g(\mathbf{x})}] d\mathbf{x}.$$

We seek to approximate an arbitrary distribution $f(\mathbf{x})$ by a normal distribution $g(\mathbf{x}) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Show that the values that minimize $D(f,g)$ are

$$\boldsymbol{\mu} = \mathbb{E}_{f(\mathbf{x})}[\mathbf{x}],$$

$$\boldsymbol{\Sigma} = \mathbb{E}_{f(\mathbf{x})}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T]$$

Given $D(f,g)$, let's first try to simplify so it's less intimidating to understand:

$$D(f,g) = \int_{-\infty}^{\infty} f(\mathbf{x}) \log[\frac{f(\mathbf{x})}{g(\mathbf{x})}] d\mathbf{x}$$

$$= \int_{-\infty}^{\infty} f(\mathbf{x}) \log(f(\mathbf{x})) d\mathbf{x} - \int_{-\infty}^{\infty} f(\mathbf{x}) \log(g(\mathbf{x})) d\mathbf{x}$$

From here, we aim to minimize D with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, which are both parameters of $g(\mathbf{x})$, so we note that we only really need to care about optimizing in terms of $g(\mathbf{x})$. Looking back at our above terms, we note that this is all in terms of the arbitrary $f(\mathbf{x})$, so there's nothing to be done here in terms of $g(\mathbf{x})$. We are subtracting the second term, so minimizing $D(f,g)$ overall means maximizing

$$\int_{-\infty}^{\infty} f(\mathbf{x}) \log(g(\mathbf{x})) d\mathbf{x}$$

which as hinted at by the expectations in terms of $f(\mathbf{x})$ (2.5 The Normal Density from Pattern Classification, by Duda, et al.), this can be translated into

$$= \mathbb{E}_{f(\mathbf{x})}[\log(g(\mathbf{x}))]$$

We substitute in the definition for the multivariate normal distribution for $g(\mathbf{x})$ (2.5.2: Multivariate Density from Pattern Classification, by Duda, et al.):

$$= \mathbb{E}_{f(\mathbf{x})}[\log(\frac{1}{(2\pi)^{k/2}|\boldsymbol{\Sigma}|^{1/2}} \exp(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T)\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}))]$$

$$= \mathbb{E}_{f(\mathbf{x})}[-\frac{k}{2}\log(2\pi) - \frac{1}{2}\log(|\boldsymbol{\Sigma}|) - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})]$$

which we now can effectively attempt to optimize with regards to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. We take the gradient of each separately and setting equal to 0 to find the corresopnding critical points for maximizing the term:

$$\nabla_{\boldsymbol{\mu}} = \frac{\partial}{\partial \boldsymbol{\mu}}\mathbb{E}_{f(\mathbf{x})}[-\frac{k}{2}\log(2\pi) - \frac{1}{2}\log(|\boldsymbol{\Sigma}|) - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})] = 0$$

$$\frac{\partial}{\partial \boldsymbol{\mu}}\mathbb{E}_{f(\mathbf{x})}[-\frac{k}{2}\log(2\pi) - \frac{1}{2}\log(|\boldsymbol{\Sigma}|) - \frac{1}{2}(\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} - \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} - \mathbf{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})] = 0$$

$$\frac{\partial}{\partial \boldsymbol{\mu}}\mathbb{E}_{f(\mathbf{x})}[-\frac{k}{2}\log(2\pi) - \frac{1}{2}\log(|\boldsymbol{\Sigma}|) - \frac{1}{2}(\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} - 2\boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})] = 0$$

$$\mathbb{E}_{f(\mathbf{x})}[0 - 0 - \frac{1}{2}(0 - 2\boldsymbol{\Sigma}^{-1}\mathbf{x} + 2\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})] = 0$$

$$\mathbb{E}_{f(\mathbf{x})}[-\frac{1}{2}(-2\boldsymbol{\Sigma}^{-1}\mathbf{x} + 2\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})] = 0$$

$$\mathbb{E}_{f(\mathbf{x})}[\boldsymbol{\Sigma}^{-1}\mathbf{x} - \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}] = 0$$

$$\mathbb{E}_{f(\mathbf{x})}[\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})] = 0$$

Since $\boldsymbol{\Sigma}$ and $\boldsymbol{\mu}$ are both not related to $f(\mathbf{x})$:

$$\boldsymbol{\Sigma}^{-1}\mathbb{E}_{f(\mathbf{x})}[\mathbf{x}] - \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} = 0$$

$$\boldsymbol{\Sigma}^{-1}\mathbb{E}_{f(\mathbf{x})}[\mathbf{x}] = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$$

$$\boxed{\boldsymbol{\mu} = \mathbb{E}_{f(\mathbf{x})}[\mathbf{x}]}$$

exactly as we sought to find. Now moving onto taking the gradient with respect to $\boldsymbol{\Sigma}$. From expanding the last term previously, we recognize that taking the function with regards to $\boldsymbol{\Sigma}$ seems a bit painful, so it is easier to derive using $\boldsymbol{\Sigma}^{-1}$ as done in 10.4.2. Case 2: All Parameters Unknown from Pattern Classification, by Duda, et al.:

$$\nabla_{\boldsymbol{\Sigma}^{-1}} = \frac{\partial}{\partial \boldsymbol{\Sigma}^{-1}}\mathbb{E}_{f(\mathbf{x})}[-\frac{k}{2}\log(2\pi) - \frac{1}{2}\log(|\boldsymbol{\Sigma}|) - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})] = 0$$

$$\frac{\partial}{\partial \boldsymbol{\Sigma}^{-1}}\mathbb{E}_{f(\mathbf{x})}[-\frac{k}{2}\log(2\pi) - \frac{1}{2}\log(|\boldsymbol{\Sigma}|) - \frac{1}{2}(\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} - \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} - \mathbf{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})] = 0$$

$$\frac{\partial}{\partial \boldsymbol{\Sigma}^{-1}}\mathbb{E}_{f(\mathbf{x})}[-\frac{k}{2}\log(2\pi) + \frac{1}{2}\log(|\boldsymbol{\Sigma}^{-1}|) - \frac{1}{2}(\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} - 2\boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})] = 0$$

$$\mathbb{E}_{f(\mathbf{x})}[0 + \frac{1}{2} \cdot \frac{\partial}{\partial \boldsymbol{\Sigma^{-1}}}(\log(|\boldsymbol{\Sigma}^{-1}|) - (\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} - 2\boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}))] = 0$$

$$\mathbb{E}_{f(\mathbf{x})}[\frac{1}{2}(|\boldsymbol{\Sigma}^{-1}|^{-T} - \frac{\partial}{\partial \boldsymbol{\Sigma^{-1}}}(\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} - 2\boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}))] = 0$$

$$\mathbb{E}_{f(\mathbf{x})}[\frac{1}{2}(\boldsymbol{\Sigma} - \frac{\partial}{\partial \boldsymbol{\Sigma^{-1}}}(\mathbf{x}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} - 2\boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}))] = 0$$

$$\mathbb{E}_{f(\mathbf{x})}[\frac{1}{2}(\boldsymbol{\Sigma} - (\mathbf{x}\mathbf{x}^T - 2\mathbf{x}\boldsymbol{\mu}^T + \boldsymbol{\mu}\boldsymbol{\mu}^T))] = 0$$

$$\mathbb{E}_{f(\mathbf{x})}[\frac{1}{2}(\boldsymbol{\Sigma} - (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T)] = 0$$

Again, as $\boldsymbol{\mu}$ is not related to $\mathbf{x}$:

$$\frac{1}{2}\boldsymbol{\Sigma} - \frac{1}{2}\mathbb{E}_{f(\mathbf{x})}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] = 0$$

$$\frac{1}{2}\boldsymbol{\Sigma} = \frac{1}{2}\mathbb{E}_{f(\mathbf{x})}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T]$$

$$\boxed{\boldsymbol{\Sigma} = \mathbb{E}_{f(\mathbf{x})}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T]}$$

exactly as we sought to find.

2.

The columns in the file `midterm_exam_data.txt` are the X and Y values of a function which is a sum of cosines with added Gaussian noise.

(a)

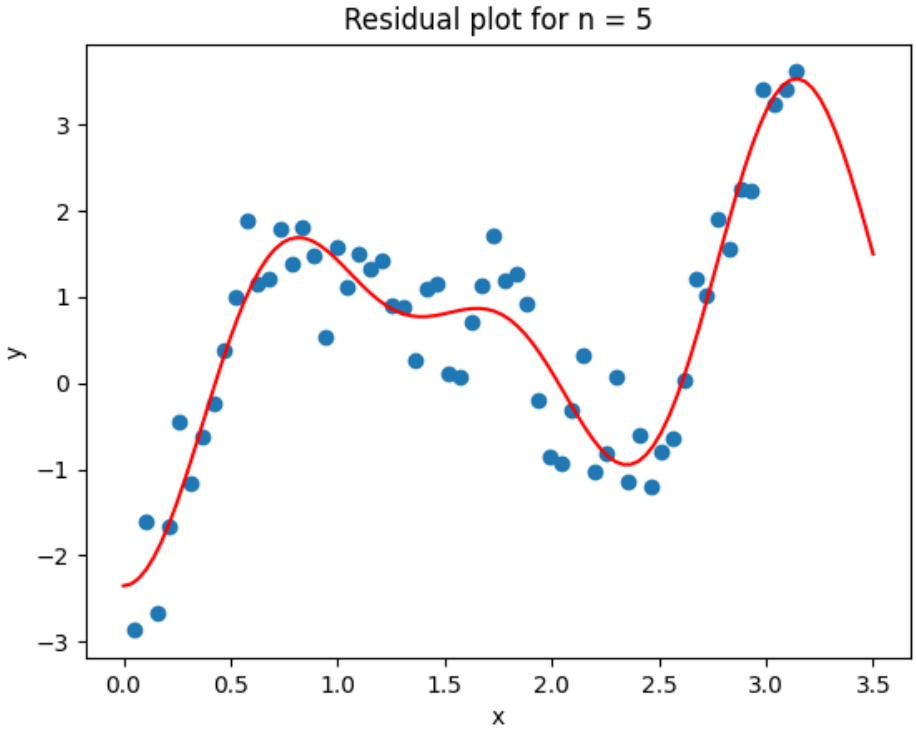
For various values of n, fit an nth order cosine series ($y = \frac{a_0}{2} + \sum_{k=1}^{n} a_k \cos kx$) to the data. What would you say is the most likely value of n?

I ran a python script and got the following output:

```
> python problem_2.py
          rss      rse  adj_r_2
n
0   115.87160  1.40140  0.00000
1   107.03410  1.35846  0.06034
```

```
2    106.96723   1.36990   0.04446
3     38.57145   0.82993   0.64929
4     37.30363   0.82356   0.65465
5     13.40291   0.49820   0.87362
6     13.23616   0.49974   0.87284
7     13.18306   0.50351   0.87091
8     13.10839   0.50698   0.86913
9     12.84362   0.50683   0.86920
10    12.78047   0.51071   0.86719
```

I chose to do this by looking at the measures used in section 3.1.3: Assessing the Accuracy of the Model from James, G. et. al, An Introduction to Statistical Learning: with Applications in R, Springer. Looking at this, the RSS seems to really improve and stabilize around $\boxed{n = 5}$ and not really improve drastically afterwards. This is the same case with the R squared and the RSE values too. I also plotted to see if this seemed sane, and it definitely seems to fit pretty well:



I got this from running the following python code:

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import statsmodels.api as sm

data = np.loadtxt("midterm_exam_data.txt")

x_data = data[:, 0]
y_data = data[:, 1]

def cos_function(x_data, n):
  """BUild the corresponding matrix by stacking 1, cos(x), etc."""
  return np.column_stack(
    [np.ones_like(x_data)] +
    [np.cos(k * x_data) for k in range(1, n + 1)]
  )

# Try for n-values up to 10
fit_results = []
result_summaries = []
for n in range(0, 11):
  fit_result = sm.OLS(y_data, cos_function(x_data, n)).fit()
  fit_results.append(fit_result)

  result_summaries.append({
    "n": n,
    "rss": fit_result.ssr,
    "rse": np.sqrt(fit_result.scale),
    "adj_r_2": fit_result.rsquared_adj,
  })
```

```
result_summary_df = pd.DataFrame(result_summaries).set_index("n").round(5)
print(result_summary_df)

# n = 5 seems reasonable, so plot and see
plt.scatter(x_data, y_data)
x_series = np.linspace(0, 3.5, 100)
plt.plot(x_series, fit_results[5].predict(cos_function(x_series, 5)), color=
plt.xlabel("x")
plt.ylabel("y")
plt.title("Residual plot for n = 5")
plt.show()
```

(b)

Estimate the level of the noise.

The output from my script was as follows:

```
Estimated noise standard deviation: 0.49819887898041026
Estimated noise variance: 0.24820212301733746
```

from the following block of code at the bottom of my python script:

```
# Print RSE as noise estimation
sigma = np.sqrt(fit_results[5].scale)
print(f"Estimated noise standard deviation: {sigma}")
print(f"Estimated noise variance: {sigma**2}")
```

The noise seems relatively low in terms of standard deviation. In addition, looking back at our table in part (a) with the RSE's seeming also relatively stable from $n = 5$ onwards, which also seems to indicate that the noise is relatively low. I chose to use RSE as a measure of noise used in section 3.1.3: Assessing the Accuracy of the Model from James, G. et. al, An Introduction to Statistical Learning: with Applications

in R, Springer

3.

Consider a Bayesian classifier, with distributions being n-dimensional multivariate Bernoulli random variables. The conditional probability for a given category is

$$P(\mathbf{x}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{x_i}(1-\theta_i)^{1-x_i},$$

and let $\mathcal{D} = \{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ be a set of k samples independently drawn from this probability density.

(a)

If $\mathbf{s} = (s_1, \ldots, s_n)^T$ is the sum of the k samples, show that

$$P(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i},$$

From the "Review of Statistics" lecture from Module 2, we know the following: With the setup of the problem with independent samples drawn from a Bernoulli distribution, this is simply the likelihood function. We can therefore just express this as the product of the probability densities:

$$P(\mathcal{D}|\boldsymbol{\theta}) = \mathcal{L}(\boldsymbol{\theta}|D) = \prod_{j=1}^{k} P(\mathbf{x}_j|\boldsymbol{\theta})$$

$$= \prod_{j=1}^{k}(\prod_{i=1}^{n} \theta_i^{x_{ij}}(1-\theta_i)^{1-x_{ij}}) = \prod_{j=1}^{k}\prod_{i=1}^{n} \theta_i^{x_{ij}}(1-\theta_i)^{1-x_{ij}}$$

$$= \prod_{i=1}^{n}\prod_{j=1}^{k} \theta_i^{x_{ij}}(1-\theta_i)^{1-x_{ij}}$$

$$= \prod_{i=1}^{n} \theta_i^{\sum_{j=1}^{k} x_{ij}}(1-\theta_i)^{\sum_{j=1}^{k} 1-x_{ij}}$$

Substituting the sum s in accordingly as given gives us exactly the desired result:

$$\boxed{P(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}$$

(b)

Assuming a uniform prior distribution for $\boldsymbol{\theta}$ show that

$$P(\boldsymbol{\theta}|\mathcal{D}) = \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i},$$

From part (a), we can find this using Bayes' thoerem (Chapter 2: Bayesian Decision Theory from Pattern Classification, by Duda, et al.):

$$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{P(\mathcal{D}|\boldsymbol{\theta}) \cdot P(\boldsymbol{\theta})}{P(\mathcal{D})}$$

$$= \frac{P(\mathcal{D}|\boldsymbol{\theta}) \cdot P(\boldsymbol{\theta})}{\int P(\mathcal{D}|\boldsymbol{\theta}) \cdot P(\boldsymbol{\theta})}$$

Substituting in our result from part (a):

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot P(\boldsymbol{\theta})}{\int \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot P(\boldsymbol{\theta})}$$

And using the assumption of the uniform distribution for $\boldsymbol{\theta}$, i.e. $P(\boldsymbol{\theta}) = 1$ for $0 \leq \theta_i \leq 1$:

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot 1}{\int \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot 1}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \int_0^1 \theta_i^{s_i}(1-\theta_i)^{k-s_i} d\theta_i}$$

We use integration by parts in the denominator:

- $u = (1-\theta_i)^{k-s_i}$
- $du = (1-\theta_i)^{k-s_i-1} \cdot (k-s_i) \cdot -1 d\theta_i = (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i$
- $v = \frac{\theta_i^{s_i+1}}{s_i+1}$
- $dv = \theta_i^{si} d\theta_i$ And substituting accordingly:

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} (1-\theta_i)^{k-s_i} \cdot \frac{\theta_i^{s_i+1}}{s_i+1}|_0^1 - \int_0^1 \frac{\theta_i^{s_i+1}}{s_i+1} \cdot (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} 0 - 0 - \int_0^1 \frac{\theta_i^{s_i+1}}{s_i+1} \cdot (s_i - k)(1-\theta_i)^{k-s_i-1} d\theta_i}$$
$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} 0 - 0 - \int_0^1 \frac{\theta_i^{s_i+1}}{s_i+1} \cdot (s_i - k)(1-\theta_i)^{k-s_i-1} d\theta_i}$$
$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} -\frac{s_i-k}{s_i+1} \int_0^1 \theta_i^{s_i+1}(1-\theta_i)^{k-s_i-1} d\theta_i}$$

We can see that from repeatedly applying integration by parts, we eventually reduce the exponent of the $1-\theta_i$ term to 0, while the exponent of the θ_i term increases until k, giving us:

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} \int_0^1 \theta_i^{k}(1-\theta_i)^{0} d\theta_i}$$
$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} \int_0^1 \theta_i^{k}(1-\theta_i)^{0} d\theta_i}$$
$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} \int_0^1 \theta_i^{k} d\theta_i}$$
$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} [\frac{\theta_i^{k+1}}{k+1}]_0^1}$$
$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} (\frac{1}{k+1} - 0)}$$
$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)!}{(s_i+1)(s_i+2)...k+1}}$$
$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)!}{(s_i+1)(s_i+2)...k+1} \cdot \frac{s_i!}{s_i!}}$$
$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)!s_i!}{(k+1)!}}$$
$$= \prod_{i=1}^{n} \frac{\theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\frac{(k-s_i)!s_i!}{(k+1)!}}$$

$$\boxed{P(\boldsymbol{\theta}|\mathcal{D}) = \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}$$

exactly as we sought to find.

(c)

Plot this density for the case $n = 1, k = 1$ and for the two resulting possibilities for s_i.

With $n = 1$, we have one sample, so our only cases are either $s = 0$ or $s = 1$:

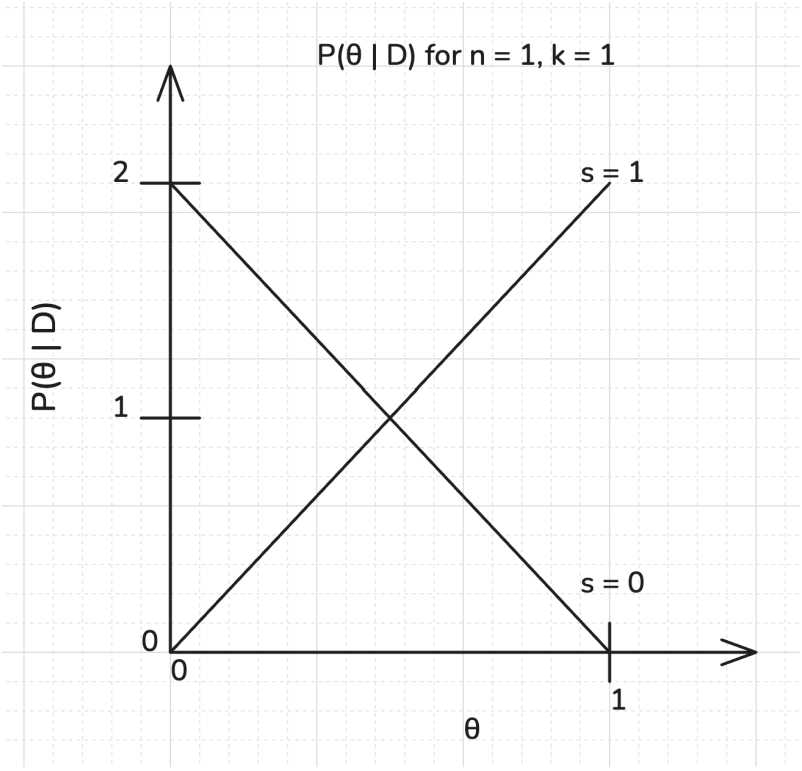
- If $s = 0$, we have

$$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{(1+1)!}{0!(1-0)!} \cdot \theta^0 \cdot (1-\theta)^{1-0} = \frac{2!}{0!1!} \cdot 1 \cdot (1-\theta)^1 = 2 - 2\theta$$

- If $s = 1$, we have

$$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{(1+1)!}{1!(1-1)!} \cdot \theta^1 \cdot (1-\theta)^{1-1} = \frac{2!}{1!0!} \cdot \theta \cdot (1-\theta)^0 = 2\theta$$

We plot this below:



(d)

Integrate the product $P(\mathbf{x}|\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathcal{D})$ over $\boldsymbol{\theta}$ to obtain

$$P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n} (\frac{s_i + 1}{k+2})^{x_i} (1 - \frac{s_i + 1}{k+2})^{1-x_i}$$

As directed, we start with

$$P(\mathbf{x}|\mathcal{D}) = \int_{\boldsymbol{\theta}} P(\mathbf{x}|\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathcal{D})d\boldsymbol{\theta}$$

(2.10: Missing and Noisy Features from Pattern Classification, by Duda, et al.) Substituting the given conditional probability and our finding from part (b):

$$= \int_{\boldsymbol{\theta}} (\prod_{i=1}^{n} \theta_i^{x_i}(1-\theta_i)^{1-x_i}) \cdot (\prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i})d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n} \int_{\boldsymbol{\theta}} \theta_i^{x_i}(1-\theta_i)^{1-x_i} \frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i}d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \int_{\boldsymbol{\theta}} \theta_i^{x_i}(1-\theta_i)^{1-x_i}\theta_i^{s_i}(1-\theta_i)^{k-s_i}d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \int_{\boldsymbol{\theta}} \theta_i^{x_i+s_i}(1-\theta_i)^{k-s_i-x_i+1}d\boldsymbol{\theta}$$

This leads us into a very similar situation as we saw in part (b) with the repeated integration by parts. In part (b), we have exponents of s_i and $k-s_i$; here, we have exponents of s_i+x_i and $k-s_i-x_i+1$, so the $1-\theta_i$ term now has 1 additional power relative to the θ_i term as it did in part (b). We just integrate by parts $k-s_i-x_i+1$ times, giving us

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\ldots(k+1)} \int_0^1 \theta_i^{k+1}(1-\theta_i)^0 d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\ldots(k+1)} [\frac{\theta_i^{k+2}}{k+2}]_0^1$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\ldots(k+1)} \cdot (\frac{1}{k+2} - 0)$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\ldots(k+2)}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\ldots(k+2)} \cdot \frac{(s_i+x_i)!}{(s_i+x_i)!}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!(s_i+x_i)!}{(k+2)!}$$

$$= \prod_{i=1}^{n} \frac{(k-s_i-x_i+1)!(s_i+x_i)!}{s_i!(k-s_i)!(k+2)}$$

$$= \prod_{i=1}^{n} \frac{(k-s_i-x_i+1)!(s_i+x_i)!}{s_i!(k-s_i)!(k+2)}$$

From this point, we can take a hint from the structure of the answer to see that we should be looking at the two different cases for x_i, that is, because it is Bernoulli, it must either be 1 or 0. Looking at these cases:

- If $x_i = 0$, we get

$$\frac{(k-s_i-0+1)!(s_i+0)!}{s_i!(k-s_i)!(k+2)} = \frac{(k-s_i+1)!s_i!}{s_i!(k-s_i)!(k+2)} = \frac{k-s_i+1}{k+2}$$

- If $x_i = 1$, we get

$$\frac{(k-s_i-1+1)!(s_i+1)!}{s_i!(k-s_i)!(k+2)} = \frac{s_i+1}{k+2}$$

And we then can express this using the exponents to account for the 0 / 1 on/off nature of the results. We therefore get

$$= \prod_{i=1}^{n} (\frac{k-s_i+1}{k+2})^{1-x_i} (\frac{s_i+1}{k+2})^{x_i}$$

$$= \prod_{i=1}^{n} (\frac{s_i+1}{k+2})^{x_i} (\frac{k-s_i+1}{k+2})^{1-x_i}$$

$$= \prod_{i=1}^{n} (\frac{s_i+1}{k+2})^{x_i} (\frac{k+2-s_i-1}{k+2})^{1-x_i}$$

$$\boxed{P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n} (\frac{s_i+1}{k+2})^{x_i} (1 - \frac{s_i+1}{k+2})^{1-x_i}}$$

exactly as we sought to find.

(e)

Compare the expressions for $P(\mathbf{x}|\mathcal{D})$ and $P(\mathbf{x}|\boldsymbol{\theta})$. What is $\hat{\boldsymbol{\theta}}$, the effective Bayesian estimate for $\boldsymbol{\theta}$?

To recap, we have

$$P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n} (\frac{s_i + 1}{k + 2})^{x_i} (1 - \frac{s_i + 1}{k + 2})^{1-x_i}$$

and

$$P(\mathbf{x}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{x_i} (1 - \theta_i)^{1-x_i}$$

We see that that θ_i directly maps to $\frac{s_i+1}{k+2}$ for both the θ_i and $1 - \theta_i$ terms, which indicates to us that $\hat{\boldsymbol{\theta}}$ is $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_n)^T$ where $\hat{\theta}_i = \frac{s_i+1}{k+2}$.